

# Smart VMS — System Architecture Submission

Smart VMS (Smart Visitor Management System)

Aligned with MoICT&NG Prototype Showcase architecture guidance

|                     |                                          |
|---------------------|------------------------------------------|
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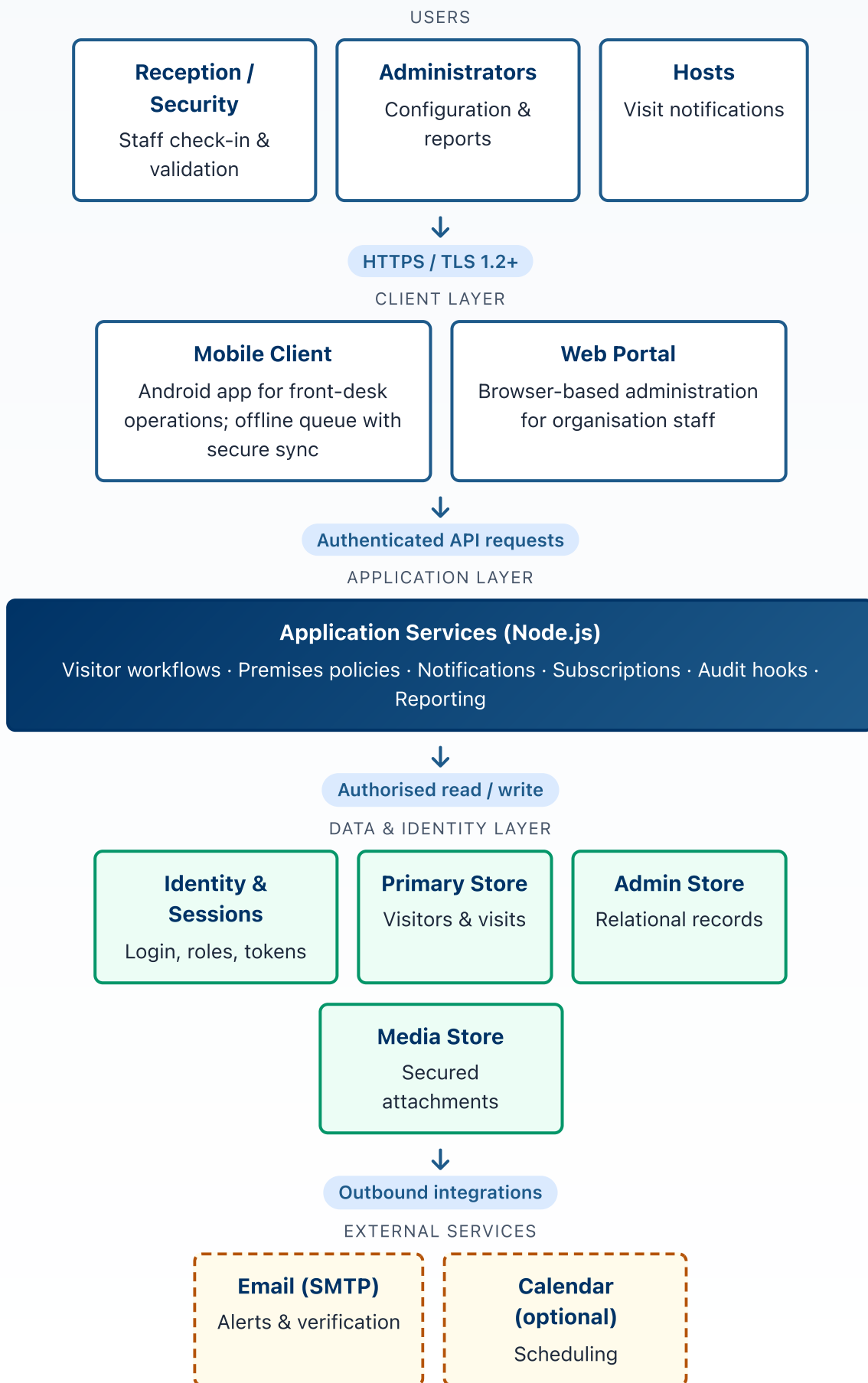
## 1. Purpose & Overview

This submission addresses the Ministry of ICT and National Guidance architecture elements: components, communication, data storage and security, user access, integrations, hosting, security controls, and data flows.

Smart VMS follows a layered architecture: client applications (mobile and web) communicate with application services over encrypted channels. Business rules enforce visitor policies, notifications, and reporting. Data is persisted in segregated stores for operational records and administration. The design supports scaling across multiple application instances and is structured for interoperability with external services through documented APIs.

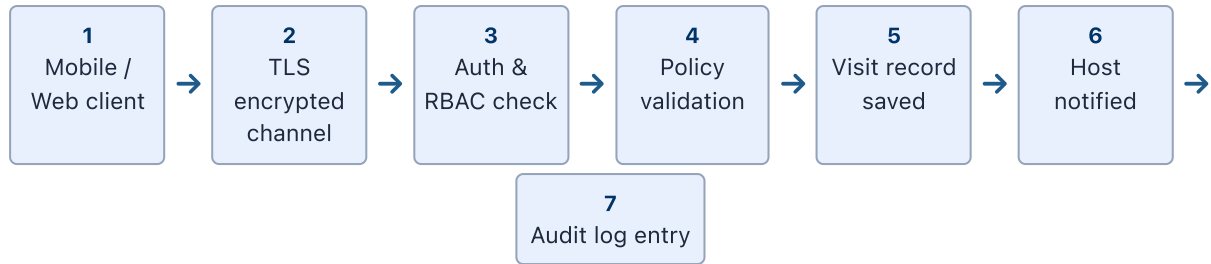
## 2. Logical Architecture Diagram

Figure 1 — Smart VMS component layers & trust boundaries



### 3. Check-in Data Flow

Primary path from reception desk to persisted visit record.



- Check-in: Staff client → HTTPS → Application services → Role validation → Visit record created → Host notification → Audit entry
- Check-out: Staff client → HTTPS → Visit record updated → Audit entry
- Administration: Web portal → HTTPS → Authenticated API → Configuration or report → Audit entry
- Premises validation: Client presents access token → Server validates policy and timing → Visit associated with premises
- Offline sync: Mobile queues events locally → On connectivity, encrypted synchronisation with conflict handling

### 4. System Components

| Component                  | Role                                                                                                             | Technology                                     |
|----------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Mobile Client              | Reception and security staff: visitor registration, check-in/out, premises validation, limited offline operation | Cross-platform mobile framework                |
| Web Administration Portal  | Organisation administrators: user management, visitor records, configuration, reports                            | Modern web application (browser-based)         |
| Application Services       | Core business logic: visits, premises, policies, subscriptions, incidents, notifications                         | Node.js modular services                       |
| Identity & Session Service | Authentication, session management, role enforcement                                                             | Industry-standard token-based access           |
| Primary Data Store         | Operational visitor and visit records, organisation configuration                                                | Document-oriented database                     |
| Administrative Data Store  | Supplementary relational records for web administration module                                                   | Relational database                            |
| Media & Document Storage   | Visitor-related attachments (e.g. capture images) with access controls                                           | Secured object storage behind application APIs |

## 5. MoICT&NG Architecture Checklist

|                                    |                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frontend / User Interface</b>   | Mobile app for reception and security (check-in, check-out, premises validation). Web portal for administrators (records, users, configuration, reports). Primary workflows: register visitor → validate access → notify host → check-out → audit/report.                                                                                                                       |
| <b>Backend / Application Layer</b> | Node.js application services host business logic. Requests authenticated and authorised per role. Stateless service instances suitable for load balancing. Request validation and audit hooks on sensitive operations.                                                                                                                                                          |
| <b>Database Layer</b>              | Categories stored: organisation profile, users and roles, premises, visitor profiles, visit events (check-in/out timestamps), audit logs, configuration. Primary operational store plus relational store for admin module. Hosting: cloud VPS (production); on-premise deployment path available. Encryption at rest on infrastructure; access restricted to application layer. |
| <b>Authentication and Access</b>   | Credential-based login with secure password handling. Session tokens validated on each request. Roles include Administrator, Manager, Receptionist, Security, and Host with least-privilege permissions. Organisation-level data isolation. MFA: planned; not required for initial deployment.                                                                                  |
| <b>APIs and Integrations</b>       | Exposes REST APIs for visitor lifecycle and administration (consumption by mobile and web clients). Consumes email (SMTP) for notifications; optional calendar connectivity for scheduling-related features. All listed integrations are in current production use.                                                                                                             |
| <b>Hosting and Infrastructure</b>  | Production: cloud virtual private server with HTTPS termination (TLS 1.2+). Reverse proxy in front of application services. On-premise deployment supported for institutions requiring local hosting.                                                                                                                                                                           |
| <b>Security Controls</b>           | TLS for all client–server traffic. Encryption at rest on database and file storage. RBAC enforced in application layer. Comprehensive audit logging (user, action, timestamp, source). Input validation and upload controls on media. Aligned with Uganda Data Protection and Privacy Act, 2019.                                                                                |
| <b>Data Flow</b>                   | Data enters at mobile/web clients → TLS → application services → authorisation check → persistence → optional notification to host. Data exits via authorised reports/exports and audit review. No public anonymous access to visitor records.                                                                                                                                  |

## 6. User Access Methods

- Android mobile application (reception and security)
- Web browser (organisation administrators)
- REST APIs (authenticated integrations — documentation in showcase repository)

## 7. Integrations (current production)

- Email (SMTP) for account verification and visit notifications
- Optional calendar integration for scheduling-related features

## 8. Security Controls

- Encrypted transport (TLS) on all client connections
- Session-based authentication with per-request validation
- Role-based access control with organisation isolation
- Audit trail for administrative and visitor-data actions
- Secure handling of uploaded media; metadata minimisation
- Rate limiting and validation on public-facing API endpoints

## 9. Scalability & Resilience

Application tier designed stateless for horizontal scaling behind a load balancer. Database tier supports replication. Suitable for institutional deployments from single premises to multi-site organisations.

## 10. Confidentiality

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